

I. Basic Properties of Numbers

- ① $3 + 4 + 8 = 15$
- ② $2 \cdot 7 \cdot 3 = 42$
- ③ $x + 3 = 7$ ($x = 4$)
- ④ $2x + 6 = 15$ ($x = 9/2$)
- ⑤ $x^2 = 9$ ($x = \pm 3$)
- ⑥ $x^2 = 5$ ($x = \pm \sqrt{5}$)
 $x^2 = -1$ ($x = \pm i$)
- ⑦ Differentiate $4x^3$: $12x^2$

Goal: Identify key properties of numbers so that statements of calculus make sense.

Addition axioms $\sim$ "+" has to make sense
if $x, y \in S$, then $x + y \in S$

(P1) Associativity: If $a, b, c \in S$, then
 $a + (b + c) = (a + b) + c$

* Consequence: " $a + b + c$ " as an expression makes sense

- All of these have the same value.

$$\begin{array}{ll} (a+b) + (c+d) & a + (b + (c+d)) \\ ((a+b) + c) + d & \text{etc.} \end{array}$$

(P2). Additive Identity

There exists an element " 0 " $\in S$,

so that for any number $a \in S$,

$$a + 0 = a = 0 + a$$

(Ex. If $S = \{1, 2, \dots, 12\}$, then $0 = 12$.)

(P3) Additive Inverse

For any $a \in S$, there exists some number $b \in S$,

so that $a + b = 0 = b + a$

" b is an additive inverse of a "

Ex. ① If $S = \mathbb{R}$, what is an additive inverse of $a = 2$?

$$A = -2$$

② What is the additive inverse of $a = 3$?

$$A = 9$$

*. Consequence

① There is a unique additive inverse for all $a \in S$.

Pf. Suppose b_1, b_2 were both add. inv of a

$$\Rightarrow \boxed{a + b_1 = 0} \quad (*)$$

$$a + b_2 = 0 = b_2 + a$$

Add b_2 to both sides of $(*)$

$$b_2 + a + b_1 = b_2 + 0$$

$$\text{By } (P_1) \quad (b_2 + a) + b_1 = b_2 + 0$$

$$\text{By } (P_3) \quad b_2 + a = 0$$

$$(P_2) \quad \begin{cases} 0 + b_1 = b_2 + 0 \\ b_1 = b_2 \end{cases}$$

②. We will denote the additive the additive inverse of a by " $-a$ ".

③. We get subtraction for free.

Ex. $S = \{1, 2, \dots, 12\}$

what is $2-5$?

$$2 + (-5) = 2 + 7 = 9.$$

Prop. In any # system satisfying (P_1, P_3) ,
then equation $x+a=b$ always has a solution.

Pf. Add $-a$ to both sides
 $x+a+(-a) = b+(-a)$

$$\text{By } (P_1) \quad x+(a+(-a)) = b+(-a)$$

$$\text{By } (P_3) \quad x+0 = b+(-a)$$

$$\text{By } (P_2) \quad x = b+(-a) = b-a$$

$\therefore$ $x = b-a \in S$ has a solu.

(P4). Commutativity If $a, b \in S$
then $a+b = b+a$ (* Try to prove).

Def Any system S satisfying (P1) - (P4) is called
an abelian group.

Multiplication (subtext: " $\cdot$ " is well-defined for $\begin{matrix} \text{any } a, b \in S \\ a, b \in S \end{matrix}$)

(P5) Associativity. For any $a, b, c \in S$, $(a \cdot b) \cdot c = a \cdot (b \cdot c)$
 $\Rightarrow a \cdot b \cdot c$ makes sense.

(P6) Multiplicative Inverse

There exists an element " 1 " $\in S$ so that for any $a \in S$,

$$a \cdot 1 = 1 \cdot a = a$$

Ex. In $S \in \mathbb{R}$, what is the multiplicative identity?
"1"

(P7). Mult. Inverse

For any number $a \neq 0$,

there is a number c so that $a \cdot c = c \cdot a = 1$.